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On the Solvation Properties of Menthol-Thymol Mixtures. A Molecular Dynamics Investigation

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Using classical molecular dynamics, we have investigated the solvation of catechol, resorcinol, hydroquinone and 1,4-benzoquinone at infinite dilution, in a series of menthol - thymol mixtures in which the molar fraction of thymol (x_{THY}) has been increased by steps of 0.1, from 0 (*pure menthol*) to 1 (*pure thymol*). The evolution of the solvation shell around the solutes reveals that when x_{THY} is increased, the average number of hydrogen bonds (HB) where the solute acts as HB acceptor (HBA) and the solvent as HB donor (HBD) increases, while the amount of HB, in which the solute acts as HBD and the solvent as HBA, decreases. Overall, the total number of HBs between the different benzenediols and the solvent decreases with an increase of x_{THY}, while for benzoquinone the total number of HB increases. This points to the fact that "acidic" or HBD molecules are better solvated in mixtures with high menthol proportion, while "basic" or HBA molecules, are better solvated in thymol rich mixtures. The results reported herein follow the same trends as experimentally reported Kamlet-Taft parameters and present insights on how the composition of these "deep eutectic" mixtures maybe tweaked in order to optimize their solvation properties.

Introduction

The use of safer solvents and auxillaries is among one of the 12 principles of green chemistry developed by P. Anastas and J. Zimmerman.[1] In this context ionic liquids (ILs) have emerged nearly 30 years ago as potential replacement of the hazardous volatile organic compounds (VOCs), commonly used as solvents in chemistry, mainly due to their negligible vapour pressure and their high thermal stability.[2] However, the years have passed and their disadvantages have also been exposed, such as for example their toxicity[3,4] and/or their non-biodegradability[5] and to date their highest drawback remains, without a doubt, their expensive production costs making their use at large scale utopic.[6]

Based on this ascertainment and hence, the need to find an alternative to ILs, deep eutectic solvent (DES), introduced by *Abbott et al.*[7] in 2002, have since emerged as a more viable alternative to ILs as a potential new class of "green" solvents, as they possess all the main advantages of the latter (non-flammable and low volatility) without having its drawbacks, as DES are easy to prepare, and are often cheap and have a low toxicity.[6] DES can be defined as an eutectic solvent whose components present enthalpic-driven negative deviations from thermodynamic ideality.[8] Generally, DES are either found to be mixtures of Lewis or Brønsted acid and base and /or formed between hydrogen bond donors (HBD) and hydrogen bond acceptor (HBA).

Certainly, the most underrated feature of these solvents is the possibility to fine tune their physico-chemical properties by modifying the composition of the solvent, which has led them to be considered as solvents that can be tailor-made. In the case of DES, the fine-tuning of the properties of the solvent can be achieved either by modifying the nature of at least one of the components of the system or by modifying the ratio between both components in the mixture. Indeed in the latter case, as shown by Alizadeh et al.,[9] no "magic" happens in the liquid phase at the specific composition of the mixture corresponding to the deep eutectic point compared to other random stoichiometries, hence the use of DES as solvents should not be limited to the fixed proportion of the deep eutectic composition, but rather the vast space of possible composition should be explored in order to maximize the solvents performance for a given application.[8] While the former case has been widely studied, studies of the latter case are far more scarce. One example of the latter has been reported by Martins *et al.*[10] which showed how the solvatochromic parameters of menthol – thymol mixtures, forming a type-V DES,[11] vary continuously in the range between pure menthol and thymol compounds. They further found that the optimal composition of the menthol-thymol extraction phase, for the selective separation of N,N-diethyl-4-nitroaniline and choranilic acid from an aqueous solution, is found at an intermediate concentration unrelated to the eutectic point.

Finding the optimal composition of a given mixture without a thorough knowledge of the interaction happening at the molecular level in these solvents can be a daunting task. Valuable information to enhance this process may come from molecular dynamic (MD) simulations, which enable for a better understanding of the interactions happening at the micro-

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scopical level inside a DES and how these vary with the exact composition of the DES. So far, many MD simulations have successfully studied the structure and dynamics of neat DES,[12,13] or their solvation properties.[14–16] Some MD simulations have even studied the effect of the modification of the nature of the components of the DES and how this could impact a given application such as for instance the extraction of polyphenols from Luma apiculate[17] or tuning the solubility of indigo dye.[18] However, to the best of our knowledge, so far no MD study has focussed on how the solvation properties of a given DES vary upon modification of the proportion of the different components in the system.

Starting from these observations we have decided to study the solvation of the 3 different isomers of benzenediol, namely catechol (1,2-benzediol), resorcinol (1,3-benzediol) and hydroquinone (1,4-benzenediol), together with 1,4-benzoquinone, the doubly oxidized form of hydroquinone (Figure 1), in various menthol-thymol mixtures going from pure menthol ($x_{THY} = 0$) to pure thymol ($x_{THY} = 1$) by steps of 0.1 in x_{THY}. The simulated systems reported in this study correspond to the solvation of these compounds at infinite dilution. The nature of the DES was chosen in order to be able to compare directly our results to the previous reported Kamlet-Taft parameters.[10] Additionally, the choice of the solutes stems from the fact that benzenediols and/or benzoquinone can be considered as simple building blocks of polyphenols, whose extraction towards DES has already largely been investigated and hence our results could give interesting insights on how to optimise these proceedings.[19,20] Furthermore, from a physico-chemical point of view, the pKa of the different benzenediol solutes are lower than that of thymol, (9.45 for catechol, 9.15 for resorcinol and 9.9 for hydroquinone), hence this molecules should be able to form stronger H-Bond (HB) which menthol than thymol. Furthermore, these 3 isomers have different dipolar moments ranging from 1.4 D in the case of hydroquinone, to 2.07 D and 2.62 D for resorcinol and catechol respectively, hence we expect to obtain insights on how the solvent adapts to changes in dipolar moment of the solute and how these changes with the composition of the solvent mixture. Finally, 1–4 benzoquinone was chosen as it is a lone hydrogen bond acceptor, which differentiates it from the benzenediol molecules which can both act as HBA and HBD.

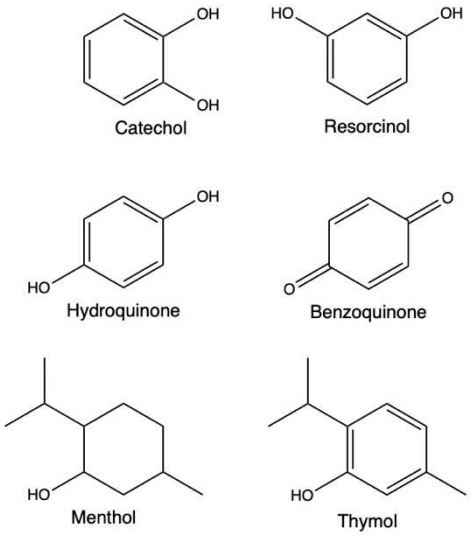


Figure 1. 2D Structure of the various solvent and solute molecules investigated.

Methods

Classical molecular dynamics were performed using the Amber 18.0[21] software where the potential energy is described by Equation (1), in which the first three terms correspond to bonded interactions (bond stretching, angle bending and torsional deformation) and the last two describe the non-bonded interactions (6–12 Lennard-Jones potential for van der Waals interactions and a coulombic term for the electrostatic interactions).

$$V(r) = \sum_{bonds} K_b(b - b_0)^2 + \sum_{angles} K_\theta(\theta - \theta_0)^2 + \sum_{dihedrals} \left(V_n/2\right)(1 + \cos[n - \gamma]) + \sum_{nonb\ ij} \left(A_{ij}/r_{ij}^{12}\right) - \left(B_{ij}/r_{ij}^{6}\right) + \left(q_i q_j/r_{ij}\right) \quad (1)$$

Force Field (FF) parameters both for the solvent (menthol and thymol molecules) as well as for the various solutes (catechol, resorcinol, hydroquinone and benzoquinone) were taken from GAFF v2.11 FF[22] with the exception of those of the OH group, which were taken from our recent work.[23] Atomic charges of the various molecules (solvent and solute) have been obtained using the RESP methodology.[24] A summary of the different FF parameters and atomic charges is given in Supporting Information. (Table S1–S4 + Figure S1 + S2).

We first performed preliminary tests on the pure menthol–thymol mixtures corresponding to the deep eutectic point to validate the choice of our FF parameters. Computed densities at different temperatures are found to be within 2% of those reported experimentally.[10] (0.9510 at 293.15 K, 0.9478 at 298.15 K, 0.9267 at 323.15 K, 0.9187 at 333.15 K and 0.9102 at 343.15 K, respectively). Furthermore, computed surface tensions of 29.3 $mN*m^{-1} \pm 1.5$ at 298.15 K, 25.9 $mN*m^{-1} \pm 1.1$ at 323.15 K and 24.8 $mN*m^{-1} \pm 0.5$ at 333.15 K, respectively, are found within 10% of those reported experimentally.[25] Hence, the use of the FF parameters allows for a good agreement with the reported experimental properties.

Each simulated system was composed of one solute molecule (catechol, resorcinol, hydroquinone or benzoquinone) and (1000–n) menthol and n thymol molecules, where n was successively increased from zero (pure menthol) to 1000 (pure thymol) by steps of 100. A summary of all the simulated system together with their main characteristics is given in Table S5.

Once the different systems were created, they were first relaxed by 5000 steps of minimization, the first 3000 steps were done using the steepest descent method while the final 2000

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were done using the conjugate gradient. They were then slowly heated to the temperature of 333.15 K at which all investigated systems have been reported to be liquid.[26] Next, in order to equilibrate the density of the system, 20 ns of MD in the NpT ensemble at a temperature of 333.15 K and a pressure of 1 bar have been performed. In order to obtain totally random initial distributions of the systems, "mixing" and "demixing" simulations were repeated three times. During the "Mixing" step the system was heated to 500 K and electrostatic interactions were scaled down by a factor 1000 for 50 ps, while during the "Demixing" step an MD of 125 ps was performed during which the system was cooled down from 500 K to 333.15 K and putting on electrostatic interactions again. The final configuration of each demixing simulation was then used as starting configuration for a "production" run. Consequently, for each system, three "production" runs of 50 ns each, in the NVT ensemble and at a temperature of T = 333.15 K were then produced.

During the various simulations, the temperature was controlled by coupling the system to an external thermal bath using the Berendsen algorithm with a relaxation time of 1.0 ps.[27] Additionally, in the NpT simulations, the pressure was controlled via the Monte Carlo barostat[28] using a relaxation time of 1.0 ps. All simulations were performed using 3D periodic conditions and then electrostatic and vdW interactions were calculated using a 12 Å cut-off. Long-range electrostatic interactions were calculated using PME (particle mesh Ewald) summation.

Structural and Energetical Analysis were either performed with our laboratory own software or using the TRAVIS software.[29] The different analyses were averaged over the 3 independent runs. In the case of the menthol - thymol mixture with $x_{THY} = 0.5$, we report the RDFs for the 3 independent runs together with the RDF averaged over these 3 runs (Figure S4) to point to fact that the systems have been sufficiently sampled. Fluctuation in the integration of these RDFs are found within a maximum of 0.05. Snapshots were taken with the VMD software.[30]

Results and Discussion

In the following section we will first describe the solvation of the different solutes (catechol, resorcinol, hydroquinone and benzoquinone) as a function of the molar fraction of thymol (x_{THY}) in the system. This will then be followed by insights on the energetics of the various systems.

Visual inspection (Figure 2) of the first solvation shell around the various solutes reveals that the former is composed by a mix of menthol and thymol molecules, whose ratio evolves with the composition of the solvent. While at low thymol concentration ($x_{THY} = 0.2$) the first solvation shell is dominated by the presence of menthol molecules, at an intermediate value ($x_{THY} = 0.5$) the first shell is found to be composed of a nearly equimolar ratio of menthol and thymol molecules and at high thymol concentration ($x_{TH} = 0.8$) an excess of thymol molecules is found in the first solvation shell of the various solutes. This is fully in line with what one would expect in the case of an ideal system. This is further corroborated by radial distribution functions (RDFs) of the centre of mass (CoM) of either menthol or thymol molecules around the CoM of the various investigated solutes (Figure S5), where a first broad peak appears in the region between roughly 3.0 and 9.0 Å, both in the case of menthol and thymol RDFs, clearly pointing to the fact that both solvent molecules actively participate in the solvation of the different solute molecules. (With the exception, of course, of the cases of pure menthol, $x_{THY} = 0$, and pure thymol, $x_{THY} = 1$, where the first solvation shell is exclusively formed by either menthol or thymol molecules.) The structure of this first peak is found to be more or less structured depending on the nature of the solute and the solvent molecule. Integration up to the minimum of this first peak in these solvent CoM – solute CoM RDFs (Table S6) shows that the number of menthol molecules around a given solute is decreasing nearly linearly when x_{THY} in the mixture is successively increased from 0.0 to 0.9. To be more accurate the average number of menthol molecules in the first shell are found to drop from 10.3 to 1.1 around catechol, from 10.2 to 1.0 around resorcinol, from 9.6 to 1.2 around hydroquinone and from 7.8 to 0.6 around benzoquinone, respectively. On the other hand, the number of thymol molecules in the first solvation shell around a given solute is growing nearly linearly when x_{THY} in the mixture is successively raised from 0.1 to 1.0. More precisely the average number of thymol molecules are seen to increase from 1.1 to 11.1 around catechol, from 0.7 to 10.9 around resorcinol, from 1.0 to 11.5 around hydroquinone and from 1.2 to 9.3 around benzoquinone, respectively.

It is worth noting that the total number of solvent molecules (menthol + thymol) in the first shell around the different solutes increases by about 10% when going from pure menthol ($x_{THY} = 0$) to pure thymol ($x_{THY} = 1$) (from 10.3 to 11.1 in the case of catechol, from 10.2 to 10.9 in the case of resorcinol, from 9.6 to 11.5 in the case of hydroquinone and from 9.0 to 9.9 in the case of benzoquinone). This results from the fact that the number of thymol molecules in the first solvation shell increases more rapidly than the number of menthol molecule decreases when x_{THY} is increased and is mainly due to two factors: i) thymol molecules are somewhat smaller than menthol molecules therefore taking up less space and ii) first shell thymol molecules may interact with each other via π–π interactions, in particular in mixture with a high x_{THY}, enabling for a more compact packing and hence further reducing the space needed in the first shell.

Hydrogen bonds (HBs) are known to play a crucial role in the structure of neat DES, hence we wanted to investigate their role in the solvation properties of these mixtures. In our prospected systems both solvent molecules (menthol and thymol) can act both as HBD or as HBA. This is also the case, with the exception of benzoquinone which can only play the role of HBA, for the different solute molecules investigated. Consequently, the two types of interactions will be considered here. We will start by discussing the case where the solvent molecules play the role of HBD and the solute that of HBA, this

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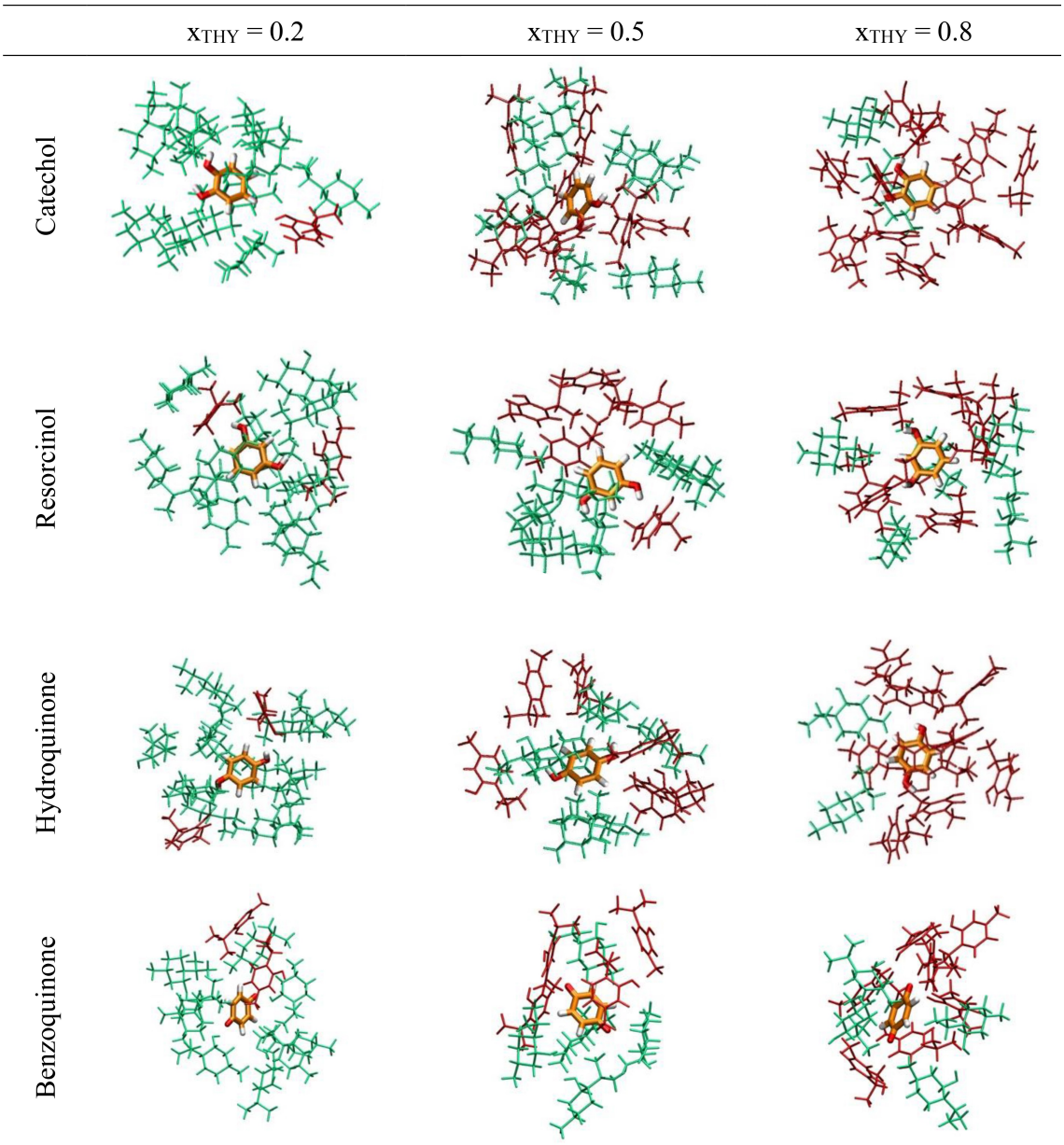


Figure 2. Final snapshot of the first solvation shell around catechol (1st line), resorcinol (2nd line), hydroquinone (3rd line) and benzoquinone (4th line) at 3 different thymol molar fractions (x_{THY} = 0.2, 0.5 and 0.8) in the solvent mixture (Menthol molecules are represented in green, thymol molecules in red).

will be followed by the description of HBs in which the solvent molecules play the role of HBA and the solute that of the HBD.

RDFs of the hydrogen atoms of the –OH group (Ho) of either menthol or thymol molecules around either the oxygen atoms of the –OH group (Oh) of catechol, resorcinol or hydroquinone or the O atoms of benzoquinone are given in Figure 3 and their main characteristics in Table S7. The presence of a first peak with a maximum around 1.95–2.15 Å in all these various RDFs points to the existence of $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ HB interactions in these mixtures. Furthermore, for a given solvent molecule, either menthol or thymol, the intensity of these peaks increases in the order catechol < resorcinol < hydroquinone < benzoquinone indicating that the HB interaction strength increases along this order. Additionally, we further observe that for a given solute molecule the intensity of the first peak is higher in the case of thymol than in the case of menthol, pointing to the fact that thymol is a stronger HBD than menthol. This is in line with the fact that the hydroxyl proton of the thymol molecule is known to be more acidic than that of menthol.[26]

Average HB distances between the solute (HBA) and the solvent (HBD), which can be derived from the position of the maximum of the first peak in the RDFs, are found to grow along the series: benzoquinone (1.97 Å ± 0.02) < hydroquinone (2.04 Å ± 0.02) < resorcinol (2.07 Å ± 0.02) < catechol (2.10 Å ± 0.04) when menthol acts as HBD and benzoquinone (1.96 Å ± 0.02) < hydroquinone (1.99 Å ± 0.02) < resorcinol (2.01 Å ± 0.02) < catechol (2.04 Å ± 0.04) when thymol plays the role of

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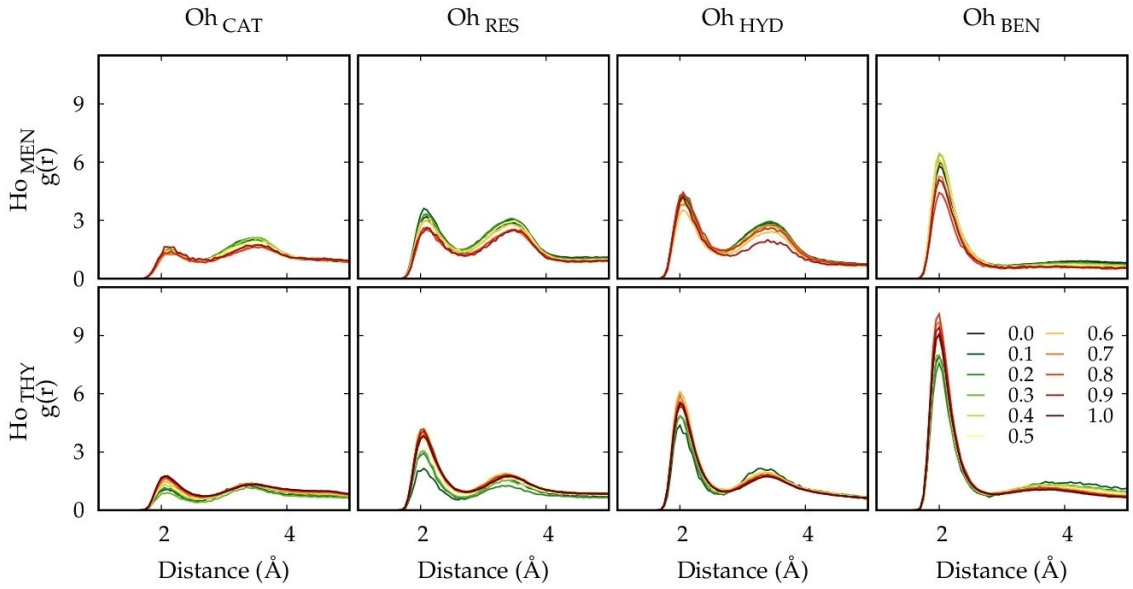


Figure 3. Radial distribution functions of $Ho_{(Solvent)}$ (Solvent = Menthol or Thymol) around the different $Oh_{(Solute)}$ (Solute = Catechol, Resorcinol, Hydroquinone or Benzoquinone) as a function of x_{THY}.

HBD, corroborating with the trends described above that suggest that the strongest $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ HB are found between benzoquinone and both solvent molecules. Conversely, the weakest $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ HBs are formed with catechol and also that the different solutes form stronger HBs with thymol than with menthol.

Integration of the first peak of the $Ho_{(MEN)}$–$Oh_{(SOLUTE)}$ RDFs is found to decay nearly linearly with x_{THY} in the range from 0.0 to 0.9 for all different solutes (Figure 4). (It should be noted that the integration corresponds to the average HB interaction of the solvent, here menthol molecules, with one -OH group of the solute, and hence should be multiplied by two to have the average HB interaction between the solute and the solvent.) In the case of catechol, the value decreases from 0.17 to 0.03, while it decreases from 0.37 to 0.04 in the case of resorcinol, from 0.45 to 0.05 in the case of hydroquinone and from 0.61 to 0.06 in the case of benzoquinone, respectively. On the contrary, integration of the first peak of the $Ho_{(THY)}$–$Oh_{(SOLUTE)}$ is found to grow linearly with x_{THY} in the range of 0.1 to 1.0. (From 0.01 to 0.25 for catechol, from 0.02 to 0.45 for resorcinol, from 0.04 to 0.58 for hydroquinone and from 0.07 to 0.86 for benzoquinone, respectively). It is further worth noting that the low values observed in the case of catechol result from the fact that the latter forms intramolecular HBs leaving less possibilities for the solvent to interact with it. Overall, the sum of both integrations, corresponding to the total number of $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ HB, is found to increase by 0.08 in the case of catechol, by 0.08 in the case of resorcinol, by 0.13 in the case hydroquinone and by 0.44 in the case of benzoquinone, when x_{THY} is increased from 0 to 1, pointing to an increasing number of $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ HB interactions at higher thymol concentration, in accordance with experimentally reported Kamlet-Taft α parameters for these menthol-thymol mixture.[10]

We now turn to the case of $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB interactions where the solvent plays the role of the HBA and the solute that of the HBD. The corresponding RDF between the Ho atoms of the catechol, resorcinol or hydroquinone molecules and the Oh atoms of menthol, respectively thymol molecules are given in Figure 5, while their main characteristics are reported in Table S8. Similarly to what was previously observed for the $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ RDFs, we observe the presence of a first peak in all the different RDFs with a maximum around 1.95–2.05 Å revealing the existence of $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB interactions in these mixtures. A first observation that can rapidly be made is that the intensity of the peaks is higher than that previously observed in the case of $Ho_{(SOLVENT)}$–$Oh_{(SOLUTE)}$ RDFs, pointing to the fact that the latter interactions are somewhat weaker than the $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB interactions. A second observation that can also easily be drawn is that the first peaks in the $Ho_{(SOLUTE)}$–$Oh_{(MEN)}$ RDFs are found to be more pronounced than their $Ho_{(SOLUTE)}$–$Oh_{(THY)}$ counterpart pointing to the fact that HB interactions should be stronger in the former case. This is further confirmed by the average HB distances, derived from the maximum of the first peak, which are found to be shorter with menthol than with thymol, (1.96 Å ± 0.01 *vs.* 2.03 Å ± 0.03 for catechol, 1.93 ± 0.01 *vs.* 1.99 ± 0.02 for resorcinol, 1.94 Å 0.02 *vs.* 2.01 Å ± 0.02 for hydroquinone, respectively), pointing to the fact that menthol is a stronger HBA than thymol and in-line with a higher basicity of the oxygen atom of the hydroxyl group in menthol than in thymol.

The integration number of the $Ho_{(SOLUTE)}$–$Oh_{(THY)}$ RDF up to its first minimum is found to increase from 0.04 to 0.32 in the case of catechol, from 0.05 to 0.50 for resorcinol and from 0.05 to 0.59 for catechol, when the molar fraction of thymol in the mixture is increased from 0 to 1, while at the same time the integration of the first peak of the $Ho_{(SOLUTE)}$–$Oh_{(MEN)}$ RDFs

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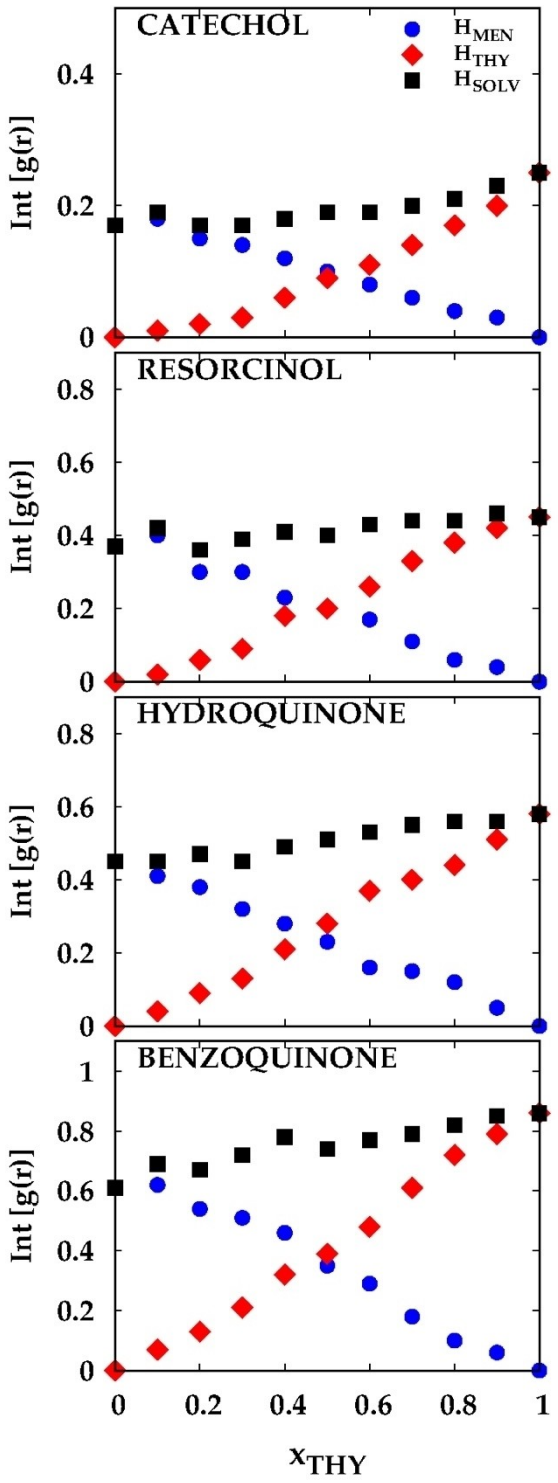


Figure 4. Integration of the first peak of the RDF in Figure 3, corresponding to the area before the first minimum, as a function of x_{THY}. (Variation of the number of HB between menthol (HBD) and the various solutes (HBA) (blue circles), variation of HB between thymol (HBD) and the various solutes (HBA) (red squares) and total sum of HB between the solvent (HBD) and the various solutes (HBA) (black squares)).

decreases from 0.59 to 0.04 in the case of catechol, from 0.90 to 0.08 in the case of resorcinol, from 0.93 to 0.05 in the case of hydroquinone. Overall, the drop in $Ho_{(SOLUTE)}$–$Oh_{(MEN)}$ HB interaction with increasing x_{THY} in the system is found to be more significant than the rise of H_{SOLUTE}–O_{THY} HB interaction leading to an overall decrease in $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB interactions (Figure 6). In particular, the number of $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB interaction is seen to drop by 0.27 for catechol, 0.40 for resorcinol and 0.43 in the case of hydroquinone, in good agreement with experimentally reported Kamlet-Taft parameters β for these mixtures.[10]

Evolution of the total HB interactions ($Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB + $Ho_{(SOLUTE)}$–$Oh_{(SOLVENT)}$ HB) between the solvent and the solute as a function of x_{THY} are given in Figure 7. They reveal two clear trends i) the number of HB interactions between the solvent and the catechol, resorcinol and hydroquinone solutes is decreasing (by 0.19, 0.32 and 0.31, respectively) when the molar fraction of thymol is increased from 0 to 1 in the system, ii) the number of HB interactions between the solvent and the benzoquinone molecule increases by 0.27 when the molar fraction of thymol is increased from 0 to 1 in the system, indicating that the catechol, resorcinol and hydroquinone molecules are better solvated in menthol rich mixtures while in contrary benzoquinone is better solvated in thymol rich mixtures.

We further computed the spatial distribution functions (SDF) of the Ho and Oh atoms of menthol or thymol around the various solutes (Figure 8) in the menthol-thymol mixture with $x_{THY} = 0.5$. In these SDF we see that in the case of menthol, the Oh density is found closer to the -OH group of the different solutes than that of Ho, pointing to the fact that in the HB interactions between menthol and the different solutes, menthol mainly acts as HBA. Inversely, in the case of thymol, the Ho density is found closer to the -OH group of the different solutes than that of Oh, pointing to the fact that in the HB interactions between thymol and the different solutes investigated, thymol essentially acts as HBD in these mixtures.

Similar conclusions can be drawn from the HB lifetimes in the menthol-thymol mixture with $x_{THY} = 0.5$ (Table 1), which show that for a given solute the longest lasting HB interactions can be found when menthol place the role of HBA, followed by those where thymol acts as HBD.

As the AMBER Force Field relies on a pairwise description of the interaction energies, the total energy of the system can be split into its different contributions, hence the solute ↔ solvent interaction energy can easily be calculated. Evolution of the latter as a function of x_{THY}, together with the menthol and thymol contribution to the solute↔solvent interaction energy is given in Figure 9. When the x_{THY} in the mixture is raised from 0 to 1, the solute↔menthol interaction is seen to decrease nearly linearly by ΔE of 25.6 kcal/mol in the case of catechol, 31.0 kcal/mol in the case of resorcinol, 31.5 kcal/mol in the case of hydroquinone and 24.0 kcal/mol in the case of benzoquinone, while the solute↔thymol interaction on the other hand is found to increase nearly linearly by ΔE of 23.4 kcal/mol in the case of catechol, 27.5 kcal/mol in the case of resorcinol, 28.5 kcal/mol in the case of hydroquinone and 29.3 kcal/mol in the case of benzoquinone. Hence in the case of catechol, resorcinol and hydroquinone the decrease in interaction energy between menthol and the solute is found to be more significant

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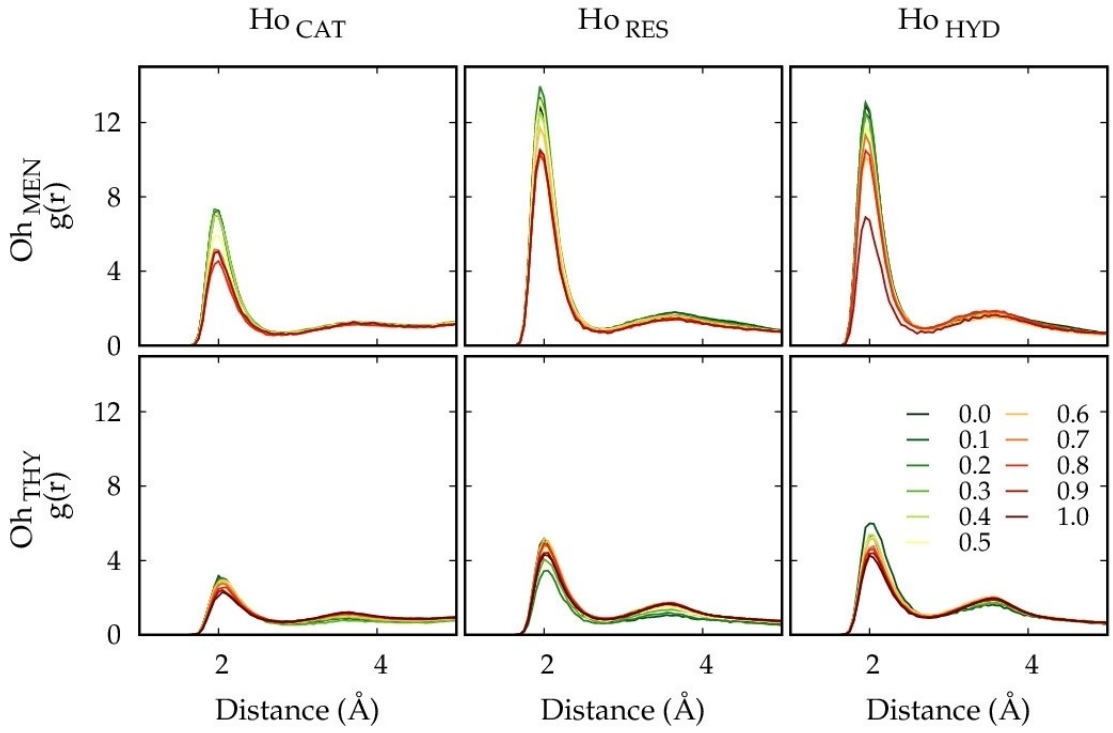


Figure 5. Radial distribution functions of $Oh_{(Solvent)}$ (Solvent = Menthol or Thymol) around the different $Ho_{(Solute)}$ (Solute = Catechol, Resorcinol, Hydroquinone or Benzoquinone) as a function of x_{THY}.

Table 1. Calculated average lifetime (ps) of H-bonding between Catechol, Resorcinol, Hydroquinone and Benzoquinone with solvent molecules menthol and thymol. Analysis run over the last 45 ns of molecular dynamics.

χ_{THY}	Catechol	Resorcinol	Hydroquinone	Benzoquinone
HBA_{SOLUTE}-HBD_{MEN}	8	24	29	9
HBD_{SOLUTE}-HBA_{MEN}	45	87	127	xxx
HBA_{SOLUTE}-HBD_{THY}	13	24	52	44
HBD_{SOLUTE}-HBA_{THY}	11	22	13	xxx

than the gain in interaction energy between thymol and the solute, leading to a decrease of the total solute↔solvent with increasing x_{THY}. ($\Delta E = 1.9$ kcal/mol for catechol, 3.5 for resorcinol and 3.0 kcal/mol for hydroquinone, respectively). On the other hand, in the case of benzoquinone we observe the opposite trend with the gain in solute↔thymol interaction energy overcompensating for the loss in solute↔solvent interaction energy, resulting in a gain in total solute↔solvent interaction energy of about 5.3 kcal/mol. Energy evolutions given here are clearly reflecting the solvation structure described above, with catechol, resorcinol and hydroquinone better solvated in menthol rich mixtures, while benzoquinone being better solvated in thymol reach mixtures.

Conclusions

Using molecular dynamics simulations, we have investigated the solvation of three different benzene-diol molecules (hydroquinone, resorcinol and catechol) as well as benzoquinone in DES mixtures composed of menthol and thymol where the molar fraction of thymol has been varied from 0 to 1 by steps of 0.1. Particular interest has been put into the interaction between the solvent and the solute.

First of all, we point to the crucial role of H-Bonds between solute and solvent in the solvation of the former. Furthermore, menthol is found to preferentially play the role of HBA in the solute–solvent HB interactions, while on the other hand thymol prefers the role of HBD in the solute–solvent HB interactions. We also show that an increase of thymol molecules in the system leads to a reduction of HB interactions between the solvent and the solute in which the solute plays the role of the HBD and the solvent plays the role of the HBA, while at the same time we observe an increase in the number of HB interaction where the solvent plays the role of HBD and the solute that of the HBA. This is total agreement with experimentally reported Kamlet-Taft parameters reported for the menthol-thymol mixtures.[10]

The total number of HB interactions between the solvent and either catechol, resorcinol and hydroquinone are found to

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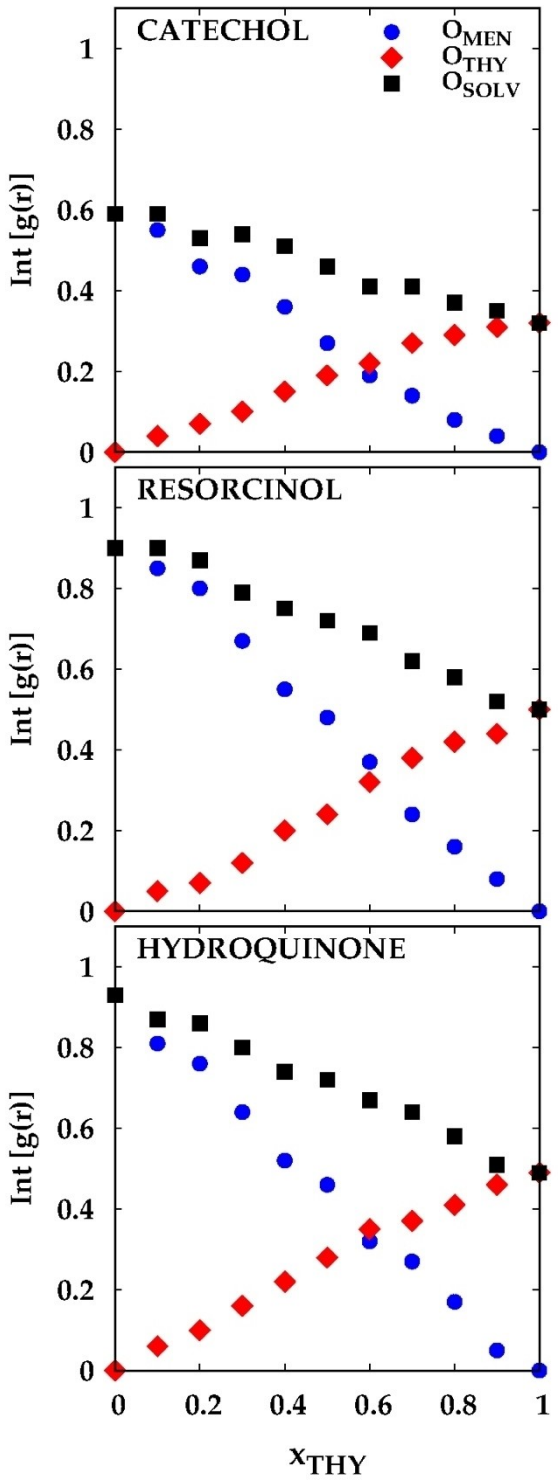


Figure 6. Integration of the first peak of the RDF in Figure 5, corresponding to the area before the first minimum, as a function of x_{THY}. (Variation of the number of HB between menthol (HBA) and the various solutes (HBB) (blue circles), variation of HB between thymol (HBA) and the various solutes (HBB) (red squares) and total sum of HB between the solvent (HBA) and the various solutes (HBD) (black squares).

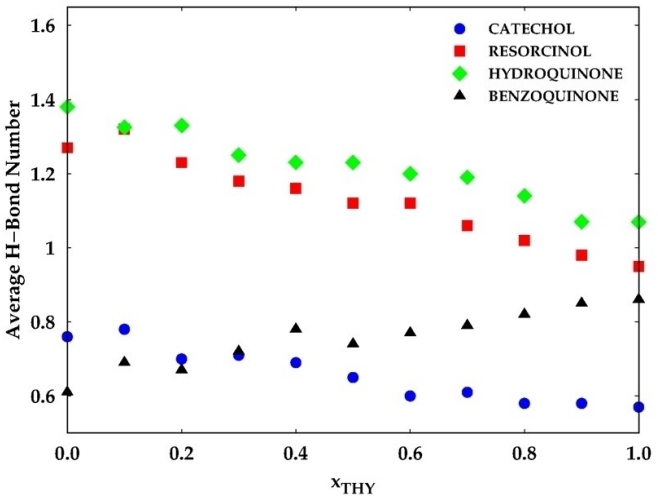


Figure 7. Total number of HB between the solvent and the various solutes as a function of x_{THY}.

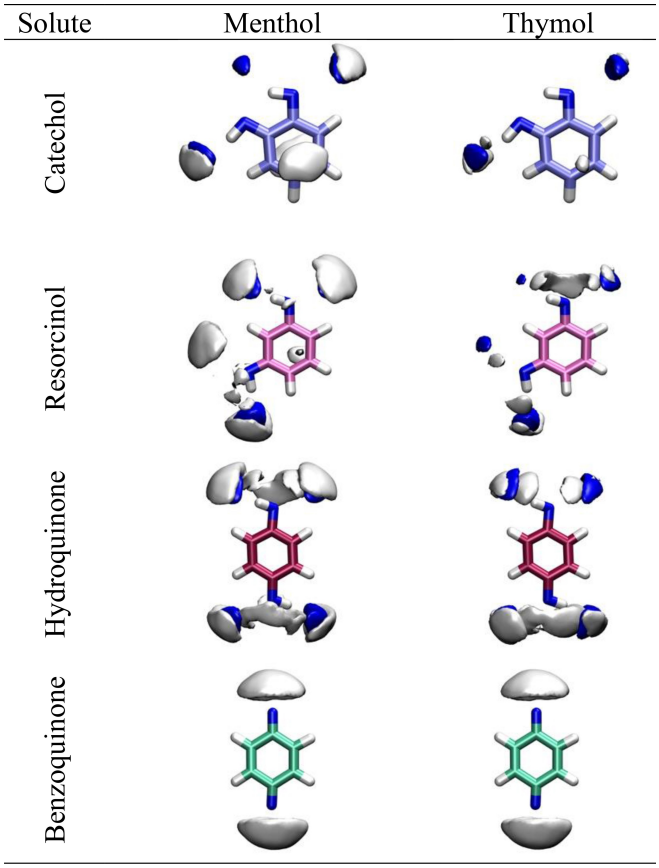


Figure 8. Menthol-Thymol mixture with $x_{THY} = 0.5$: Spatial Distribution Functions of the menthol (1st column) and thymol (2nd column) around the different solutes. (Ho(Solvent) density (Isovalue = 8.5) in white, Oh(Solvent) density (Isovalue = 15.0) in blue).

decrease when the x_{THY} is increased in the mixture, while on the other hand the total number of HB interactions between the solvent and benzoquinone are found to increase. Energy analysis shows that the catechol interacts less strongly with the solvent then resorcinol and hydroquinone, which is also related to the fact catechol, prefers to form intramolecular H-Bonds. Furthermore, solvent-solute interaction energies are seen to

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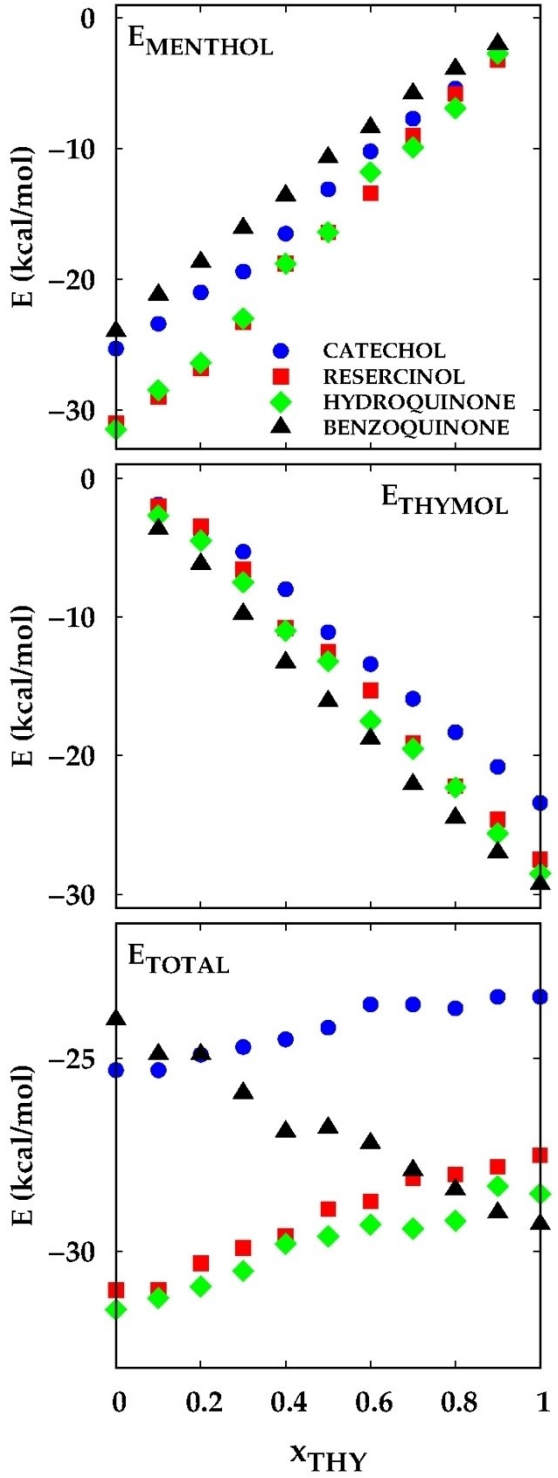


Figure 9. Solute↔menthol–(top), solute↔thymol–(middle) and total solute ↔solvent (bottom) interaction energies in kcal/mol as a function of x_{THY}.

slightly increase for all three solutes as the molar fraction of thymol is increased.

Analysis of the interaction energy between the various solutes reveal the same trends as reported. Further valuable insight into the solvation properties of these mixtures could certainly be obtained via the computation of the free energy of solvation ΔG_{SOLV}. These are however not trivial to obtain due to the high viscosity of these liquids and are currently beyond the scope of this article. Research in this direction is however currently ongoing in our laboratory.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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